

Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

Final Research Project Report

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Abstract

Irish banks process a fast-growing volume of digital transactions. Rule-based security controls struggle to keep up with it. This project evaluates how Big Data Analytics can detect and prevent cyber threats and fraud in banking. It compares supervised machine learning models on a public, labelled dataset of bank transactions (charanmaik, 2025). The dataset has 33,611 records, and 42.3% of them are flagged suspicious.

Following a quantitative experimental design, the study builds one reproducible pipeline. The pipeline covers data cleaning, feature engineering, stratified 80:20 splitting, 5-fold cross-validated hyper-parameter tuning and statistical comparison. It compares the three models in the proposal, Decision Tree, Random Forest and a Support Vector Machine. Logistic Regression, gradient boosting and a transparent two-condition rule serve as baselines.

On the held-out test set, the Decision Tree and the Random Forest perform the same: F1 is 0.996, with a 95% bootstrap confidence interval of 0.994 to 0.998. Recall is 0.9996 and ROC-AUC is 0.997 to 0.998. Both are significantly better than the SVM (F1 0.989) and Logistic Regression (F1 0.991), with McNemar $p < 0.001$. The SVM is also the slowest model, taking about 80 times longer than the Decision Tree to train.

A set of robustness experiments puts these near-perfect scores in context:

- A two-condition rule learnt from the training data reaches F1 0.995: *merchant has a suspicious history* and *amount > 197,827*. The labels therefore follow a nearly deterministic pattern.
- Removing merchant information lowers F1 to 0.80.
- For merchants that the model never saw in training, F1 falls to 0.86 to 0.88.

Big Data Analytics works best in this setting as a way to find and then operationalise simple, auditable risk patterns. Black-box accuracy on its own adds little here. The report ends with recommendations for Irish banks that link these findings to GDPR, DORA and the EU AI Act.

Keywords: Big Data Analytics, cybersecurity, banking, fraud detection, machine learning, explainability, Ireland

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1 Introduction

1.1 Research background

Online, mobile and instant payments have changed Irish retail banking. Faster and more convenient services have also widened the attack surface. Phishing, account takeover, malware, data breaches and authorised-push-payment fraud are now everyday risks for banks and their customers.

Cybercrime was projected to cost the world economy about \$10.5 trillion a year by 2025 (Cybersecurity Ventures, 2020). In Ireland, almost 90% of companies reported disruption or financial loss from cyberattacks in 2025 (The Irish Times, 2025). Banks hold large amounts of money and sensitive personal data, so they are frequent targets. When they are breached, they suffer financial loss and regulatory sanctions. They also lose customer trust.

Big Data Analytics (BDA) means collecting, storing and analysing large, fast-moving and varied data. It has become a core defensive capability. Platforms such as Hadoop and Spark (Zaharia et al., 2016), used together with machine learning (ML) libraries in Python (Pedregosa et al., 2011; Niazi, 2024), let banks score every transaction. These tools can find anomalous behaviour and adapt to new attack patterns faster than static rule sets (Buczak and Guven, 2016; Ofoegbu et al., 2023).

1.2 Problem statement

Traditional signature-based and rule-based controls are good at catching known, simple attacks. They struggle with the volume of modern transaction streams and with sophisticated or evolving fraud. The literature shows that ML-based detection works well on generic cybersecurity benchmarks (Ahmad et al., 2021; Hozouri et al., 2025). There is much less evidence on the following two points:

- (i) how established, interpretable ML models compare on *banking transaction* data;
- (ii) what that means for banks that operate under Irish and EU regulation.

1.3 Aim, objectives and research questions

Aim. To analyse the role of Big Data Analytics in detecting and preventing cyber threats in banking organisations in Ireland.

Objectives.

1. Identify how BDA is used to detect current cyber threats faced by banking organisations.
2. Evaluate the effectiveness of BDA techniques, namely machine learning and predictive analytics, for identifying and mitigating cyber threats.
3. Identify the challenges of implementing BDA in banking institutions.
4. Recommend strategies for successful implementation of BDA in Irish banks.

Research questions.

- RQ1** How is Big Data Analytics used to detect and prevent cyber threats in banking organisations in Ireland?
- RQ2** How effective are BDA-based techniques, such as machine learning and predictive analytics, at identifying and mitigating cyber threats?
- RQ3** What challenges do banking institutions face when implementing BDA?
- RQ4** Which strategies can banking institutions adopt to implement BDA successfully?

1.4 Contributions

1. A fully reproducible, open-source pipeline covering EDA, feature engineering, tuning, evaluation and explainability. It compares Decision Tree, Random Forest and SVM against three reference models on a public banking dataset.
2. A statistically grounded comparison: 5-fold cross-validation, bootstrap confidence intervals and McNemar tests.
3. A robustness analysis that the proposal did not plan: rule baseline, feature ablation, re-balancing, temporal split and unseen-merchant generalisation. It shows *why* the models score so highly and where they would fail.
4. Practical, regulation-aware recommendations for Irish banks.

1.5 Structure of the report

- Section 2 reviews related work.
- Section 3 describes the methodology.
- Section 4 presents the design and Section 5 the implementation.
- Section 6 reports the results.
- Section 7 discusses them against the research questions.
- Section 8 concludes.

2 Related Work

2.1 Big Data Analytics in cybersecurity

Buczak and Guven (2016) surveyed data-mining and ML methods for intrusion detection. They showed that hidden patterns in large security datasets support better detection than hand-written signatures. Martínez Torres et al. (2019) argued that ML reduces reliance on rule-based approaches by automating threat detection. They also warned that results depend on good data and features.

Ofoegbu et al. (2023) described how streaming BDA platforms combined with ML enable real-time threat detection in high-volume environments. Zaharia et al. (2016) set out the distributed-computing engine that makes such workloads practical. Kumar et al. (2025) tested ML models on 500,000 security incidents and reported 96.8% accuracy for deep learning, 94.1% for Random Forest and 92.3% for SVM, with 35% fewer false positives when big data analytics was combined with blockchain. Kokogho et al. (2025) reviewed how predictive modelling and behaviour analytics support cyber risk management in fintech.

Banking brings its own constraints: strict latency limits, heavy regulation, and a high cost

for false alarms. Sathupadi et al. (2025) addressed these directly with BankNet, which combines Apache Spark and Kafka with a BiLSTM model to score internet banking transactions in real time. It reached 98.5% fraud detection accuracy at up to 1000 Mbps. It is the closest example of a big data architecture built for banking, and it guides the design side of this project.

2.2 Machine learning for cyber threat detection

Early deep-learning IDS work (Javaid et al., 2016; Vinayakumar et al., 2019) and hybrid CNN-LSTM designs (Khan et al., 2019) showed accuracy gains over classical IDS. The systematic study by Ahmad et al. (2021) found that ML and deep learning beat conventional security strategies consistently. It also listed recurring problems: class imbalance, computational cost and model maintenance.

Hozouri et al. (2025) reviewed a large body of IDS research, covering signature-, anomaly- and hybrid-based detection. They named adversarial robustness, real-time latency, scalability, explainability (SHAP, LIME) and imbalanced data as open challenges.

Zhang (2025) built an ensemble of deep networks on the BoT-IoT dataset. The ensemble reached 99.985% accuracy and 99.54% F1. These results are impressive, but they come from network-traffic benchmarks, not financial transactions.

2.3 Intrusion detection systems and datasets

Ring et al. (2019) and Ferrag et al. (2020) showed that published detection performance depends heavily on the dataset. Many benchmark datasets are dated, synthetic or unrepresentative, and models tuned on them may not transfer to real deployments. This point is central to the present study. Section 6 tests directly how far performance on a public transaction dataset reflects a learnable, general signal rather than dataset artefacts.

Privacy is another IDS concern. Laddi et al. (2025) proposed a federated-learning IDS in which organisations train collaboratively without pooling raw data. It reached 94.16% accuracy and 93.82% F1. This design is attractive under GDPR (European Parliament and Council of the European Union, 2016).

Pospíchal et al. (2026) show how a Random Forest IDS should be evaluated carefully. On UNSW-NB15 they combined mutual-information feature selection, randomised hyperparameter search and SMOTE, and reported F1 between 0.93 and 0.98 across five attack types. Their protocol of tuning on training data only and reporting per-class F1 is followed in this study.

2.4 Cyber threat intelligence and predictive analytics

Montasari et al. (2020) argued that AI and ML turn large volumes of security telemetry into actionable cyber threat intelligence. This supports proactive rather than reactive defence.

Padmanaban et al. (2026) proposed a banking-specific framework, the Automated Cyber Threat Detection and Prevention (ACTP) tool. It combines reinforcement learning, ethical-hacking simulation and adversarial training. The tool reported 88.2% accuracy

and a 5.1% false-positive rate. This highlights that attackers adapt, and that robustness matters as much as accuracy on a static test set. Kovačević et al. (2025) make the same point conceptually, describing data poisoning and evasion attacks against banking AI and the dual use of AI by attackers.

Eshra et al. (2025) studied threat intelligence in financial institutions and reported that more mature threat intelligence was associated with 33.7% lower breach likelihood and 28% faster incident response. Ullah et al. (2026) proposed a risk assessment framework for US banks with a hybrid XGBoost and LSTM model (96.4% accuracy, ROC-AUC 0.982), although on simulated data.

2.5 Big Data Analytics in banking security and fraud detection

Hasan et al. (2023) reviewed Big Data in banking operations. They found clear benefits for fraud monitoring and risk management, along with persistent concerns about privacy, regulation and governance. The systematic review of ML-based financial fraud detection by Ali et al. (2022) covered 93 articles and found that credit card fraud is the most studied fraud type and that SVM and artificial neural networks are the most popular algorithms.

Recent banking studies point to tree ensembles. Sanda et al. (2026) compared Logistic Regression, Random Forest, SVM, gradient boosting and a neural network on a Kaggle banking fraud dataset with SMOTE and a 70:15:15 split; Random Forest was best with 97.2% accuracy and 98.4% AUC. Dhammayogi and Pramatha (2026) compared Random Forest, XGBoost and SVM under extreme imbalance and found ROC-AUC of 0.992, 0.995 and 0.935 respectively, with SHAP showing transaction frequency and amount spikes as the key features. Baisholan et al. (2025) combined Random Forest and XGBoost with threshold tuning and SHAP on credit card data, reaching 95% recall and 97% AUC-PR.

Naeem et al. (2026) compared rule-based, ML and AI-driven fraud detection in digital banking. In their simulated set-ups, gradient boosting (XGBoost) scored best, ahead of Random Forest, SVM and Logistic Regression. They stressed explainability for regulators. Their results are based on a *theoretical simulation* rather than real transactions, which reinforces the need for empirical comparisons on actual transaction data.

2.6 Research gap

Table 1 summarises the most relevant studies. Three gaps motivate this project:

1. few studies compare interpretable classical models (DT, RF, SVM) on *banking transaction* data using rigorous statistical tests;
2. almost none check whether very high reported scores survive tests aimed at dataset artefacts, such as ablation, temporal splits or unseen merchants;
3. Irish or EU banking regulation is rarely linked to the technical results.

Table 1: Summary of key related work.

Study	Domain / data	Method	Key limitation for this project
Buczak and Guven (2016)	Generic IDS	Survey of data mining / ML	Not banking-specific
Ahmad et al. (2021)	Network IDS	ML vs DL systematic study	Imbalance, maintenance cost
Hozouri et al. (2025)	IDS benchmarks	Survey (ML, DL, XAI)	No banking focus
Zhang (2025)	BoT-IoT	DL ensemble + SMOTE	Network traffic, not transactions
Laddi et al. (2025)	Network IDS	Federated ML	No banking data
Sathupadi et al. (2025)	Internet banking	BiLSTM with Spark and Kafka	Deep model, hard to audit
Pospíchal et al. (2026)	UNSW-NB15	RF, feature selection, SMOTE	Network traffic, not transactions
Sanda et al. (2026)	Banking fraud	LR, RF, SVM, GB, NN	Accuracy-led reporting
Dhammayogi and Pramatha (2026)	Digital banking	RF, XGBoost, SVM, SHAP	No rule baseline
Padmanaban et al. (2026)	Banking attacks	Adversarial ML and RL	Lower accuracy (88.2%)
Ali et al. (2022)	Financial fraud	Systematic review	No new experiments
Naeem et al. (2026)	Digital banking	RF, XGB, SVM, LR	Simulated results only

3 Research Methodology

3.1 Research design

The study uses a *quantitative experimental* design on secondary data. The unit of analysis is a single bank transaction and the outcome is the binary label `isSuspicious`. Models are the experimental treatments. Performance metrics on unseen data are the dependent variables. All randomness is controlled by a fixed seed (42), so every result can be regenerated exactly.

3.2 Dataset

The *Bank Transaction Records along Suspicious Flags* dataset (charanmaik, 2025) was downloaded with the Kaggle API. It contains 33,611 transactions dated 16 December 2022 to 12 January 2025. There are six raw fields:

- `date`;
- `description`, a narrative such as `NETTXN:AMAZON` or `CRYPTOEXCHANGEBINANCE`;
- `debit`, `credit` and `balance`;
- the label `isSuspicious`.

There are no missing values and no duplicate rows. The 43 distinct descriptions fall into 10 transaction channels. 14,203 transactions (42.3%) are suspicious, so the classes are

only mildly imbalanced (Figure 2a).

The dataset is public, anonymised and synthetic in style: account numbers are clearly placeholders. It contains no data about real individuals.

3.3 Pre-processing and feature engineering

Table 2 lists the engineered features. Numeric features are standardised and categorical features are one-hot encoded. Both steps run inside a scikit-learn `Pipeline`, so all transformations are fitted on the training folds only. This prevents information leaking from the test data.

Table 2: Engineered features.

Feature(s)	Definition / rationale
<code>amount, log_amount</code>	debit + credit and its $\log(1 + x)$; transaction size is a classic fraud signal.
<code>is_debit</code>	Direction of money flow.
<code>balance, log_balance</code>	Account balance after the transaction (signed log).
<code>amount_to_balance,</code> <code>negative_balance</code>	Size relative to the account and overdraft indicator.
<code>year, month, day,</code> <code>dayofweek, is_weekend</code>	Calendar context.
<code>channel</code>	Prefix of the description (e.g. NETTXN, PCA, IMPS, CRYPTOEXCHANGE).
<code>description</code>	Full merchant / narrative (43 levels).

3.4 Models

Three models come from the proposal:

- **Decision Tree** (CART) (Breiman et al., 1984): interpretable if-then rules.
- **Random Forest** (Breiman, 2001): a bagged ensemble of 300 trees.
- **Support Vector Machine** with an RBF kernel (Cortes and Vapnik, 1995).

Three reference models put these results in context:

- **Logistic Regression**, a linear baseline;
- **Histogram Gradient Boosting** (Friedman, 2001), a state-of-the-art tabular learner;
- a **Rule baseline**, which flags a transaction when both hold: its description has any suspicious history in the training data, and its amount exceeds a threshold chosen to maximise training F1.

The rule baseline mimics a traditional, analyst-written control. It shows how much the ML models add beyond it.

Class imbalance is handled with `class_weight="balanced"`. Section 6.5 compares this with SMOTE (Chawla et al., 2002) and with no re-balancing.

3.5 Evaluation protocol

1. **Hold-out split.** Stratified 80:20 split: 26,888 training and 6,723 test transactions.

2. **Tuning.** Grid search with 5-fold stratified CV on the training set, optimising F1. The grids are:
 - Decision Tree: depth $\{5, 10, 20, \infty\}$ and minimum leaf size $\{1, 5, 20\}$;
 - Random Forest: depth $\{10, 20, \infty\}$ and minimum leaf size $\{1, 5\}$;
 - SVM: $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01\}$.
3. **Metrics.** Accuracy, precision, recall, F1, ROC-AUC, PR-AUC (Saito and Rehmsmeier, 2015) and the confusion matrix. A 1,000-sample bootstrap gives the 95% CI for test F1.
4. **Significance.** Exact McNemar tests between the best model and every other model (Dietterich, 1998).
5. **Robustness.**
 - feature-group ablation;
 - re-balancing strategies;
 - a *temporal* split: train before 17 March 2024, test on or after that date;
 - *unseen-merchant* generalisation: 5-fold **GroupKFold** over descriptions, without the description feature.
6. **Explainability and operations.**
 - Random Forest impurity importance and SHAP values (Lundberg and Lee, 2017);
 - decision-threshold trade-off;
 - training time as data size grows, and scoring latency per transaction.

3.6 Ethical considerations

The research uses only public, anonymised, secondary data, so no human participants were involved and no informed consent was needed. The data is used only for academic purposes, and no attempt is made to identify people or organisations.

Model bias and fairness were considered. The label is only mildly imbalanced, and re-balancing is tested explicitly. The limitations of synthetic data for real-world claims are stated openly in Section 7.5. Any production use would require a Data Protection Impact Assessment under GDPR Article 35 (European Parliament and Council of the European Union, 2016). It would also require governance of high-risk AI under the EU AI Act (European Parliament and Council of the European Union, 2024).

4 Design Specification

Figure 1 shows the analytics architecture. The experimental core (solid boxes) is implemented in this project. The dashed boxes show how the same trained pipeline would sit inside a bank’s big-data stack:

- transactions stream in through Kafka or Spark;
- each one is scored in real time by the serialised model;
- alerts go to the fraud or SOC team;
- analyst feedback is fed back as new labels.

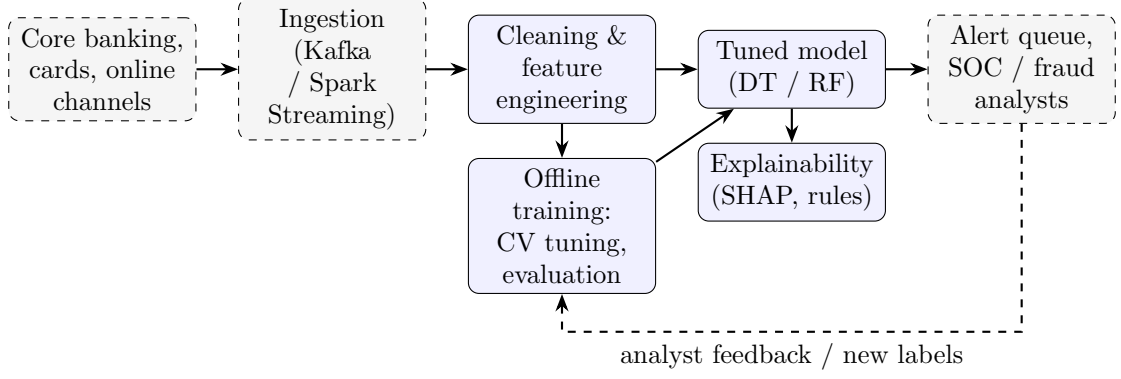


Figure 1: Proposed Big Data Analytics architecture. Solid boxes are implemented; dashed boxes show deployment context.

5 Implementation

The work was implemented in Python 3 with pandas, NumPy, scikit-learn 1.5 (Pedregosa et al., 2011), imbalanced-learn, statsmodels, SHAP, Matplotlib and Seaborn. The project repository is organised as follows:

- `src/pipeline.py`: the complete, single-command experiment (EDA → tuning → evaluation → robustness → figures). It takes about four minutes on a 15-core laptop.
- `notebooks/analysis.ipynb`: an executed Jupyter notebook that reruns the pipeline and shows every table and figure.
- `results/`: machine-readable outputs (CSV/JSON) for every table in this report.
- `figures/`: all figures.

Every pre-processing step sits inside the model pipeline, which prevents leakage from the test data. Grid search, cross-validation and Random Forest training run in parallel on all CPU cores (`n_jobs=-1`). This is the single-node equivalent of the data-parallel execution that Spark provides on a cluster.

6 Evaluation

6.1 Exploratory data analysis

Figure 2 summarises the data. Suspicious activity is concentrated in a few channels:

- crypto-exchange purchases: 71.8% suspicious;
- online (NETTXN) payments such as travel booking and e-commerce: 61.4%;
- card (PCA) purchases at online retailers.

ATM withdrawals, cash deposits, EMI repayments, IMPS transfers, refunds and salary credits are never flagged. Within the risky channels, suspicious transactions are much larger: every transaction below roughly 200,000 currency units is normal. The suspicious share also varies over time. It peaks in 2023 at 50.2%, against 19.5% in 2024.

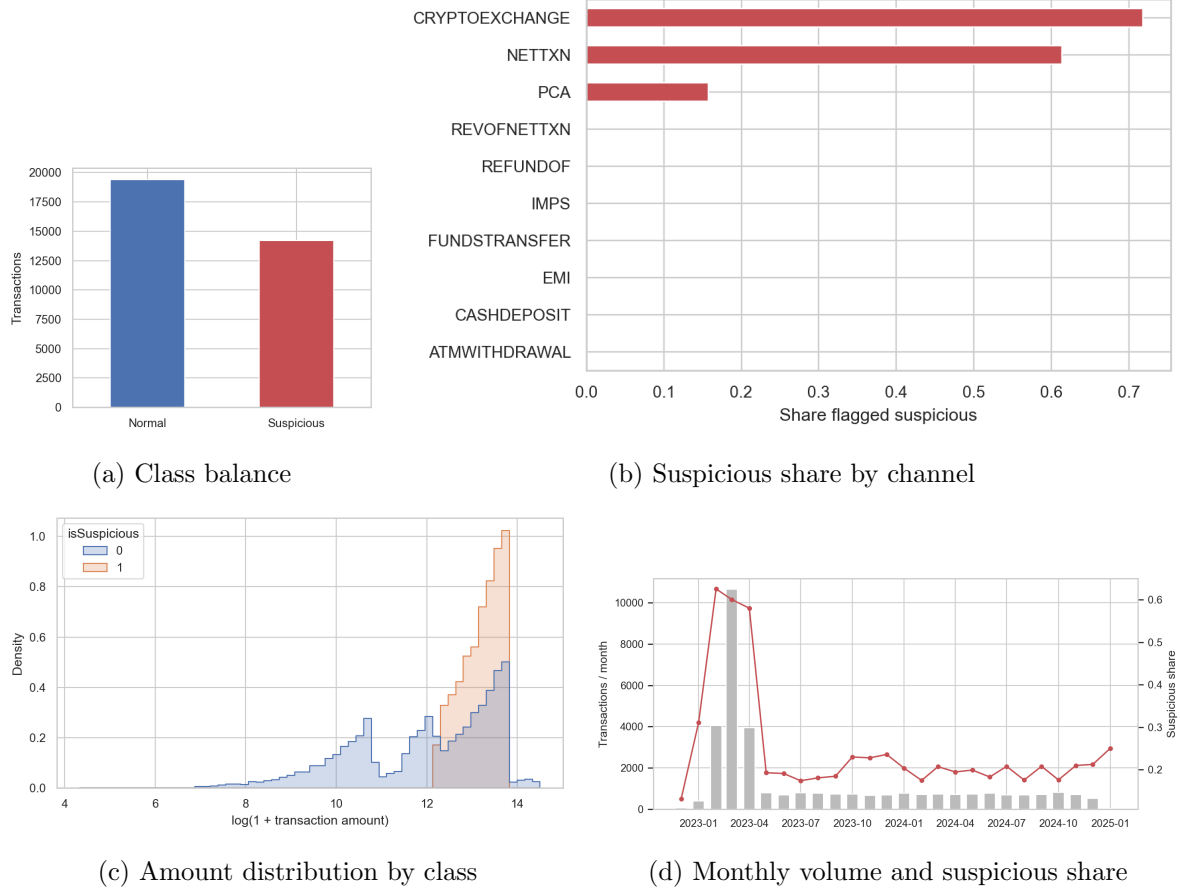


Figure 2: Exploratory analysis of the 33,611 transactions.

6.2 Model comparison

Table 3 shows the selected hyper-parameters. Table 4 and Figure 3 report performance on the held-out test set.

Table 3: Selected hyper-parameters (5-fold CV, F1).

Model	Best parameters	CV F1
Decision Tree	max_depth = 10, min_samples_leaf = 5	0.9963
Random Forest	max_depth = 20, min_samples_leaf = 1	0.9963
SVM (RBF)	$C = 10$, $\gamma = \text{scale}$	0.9874

Table 4: Test-set performance ($n = 6,723$; 2,841 suspicious). Latency is per transaction, measured for batch scoring. Rule-baseline AUCs are computed from hard labels.

Model	Acc.	Prec.	Rec.	F1 [95% CI]	ROC-AUC	PR-AUC	FP	FN
Decision Tree	0.9966	0.9923	0.9996	0.9960 [0.994, 0.998]	0.9976	0.9938	22	1
Random Forest	0.9966	0.9923	0.9996	0.9960 [0.994, 0.998]	0.9971	0.9938	22	1
Gradient Boosting	0.9964	0.9916	1.0000	0.9958 [0.994, 0.997]	0.9987	0.9977	24	0
Rule baseline	0.9958	0.9902	1.0000	0.9951 [0.993, 0.997]	0.9964	0.9902	28	0
Logistic Regression	0.9924	0.9830	0.9993	0.9911 [0.989, 0.994]	0.9965	0.9913	49	2
SVM (RBF)	0.9909	0.9857	0.9930	0.9893 [0.987, 0.992]	0.9965	0.9911	41	20

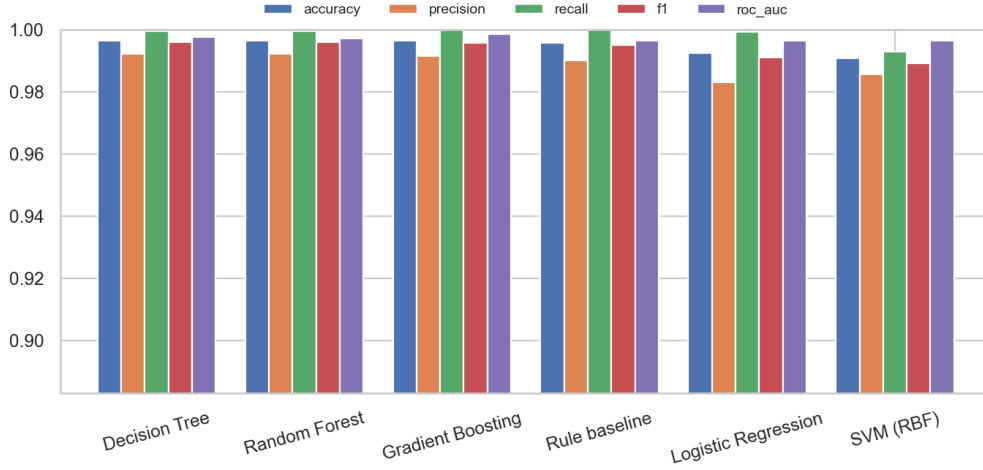


Figure 3: Test-set metrics by model (note the truncated y-axis).

All models are highly accurate. The two tree-based models from the proposal make exactly the same predictions on the test set: 22 false alarms and a single missed suspicious transaction out of 2,841. Gradient boosting misses none but raises two extra false alarms, and it has the best ranking quality (ROC-AUC 0.9987). The SVM is clearly the weakest of the proposal’s models. It misses 20 suspicious transactions, twenty times as many as the tree models.

Figure 4 shows the ROC and precision-recall curves, and Figure 5 the confusion matrices.

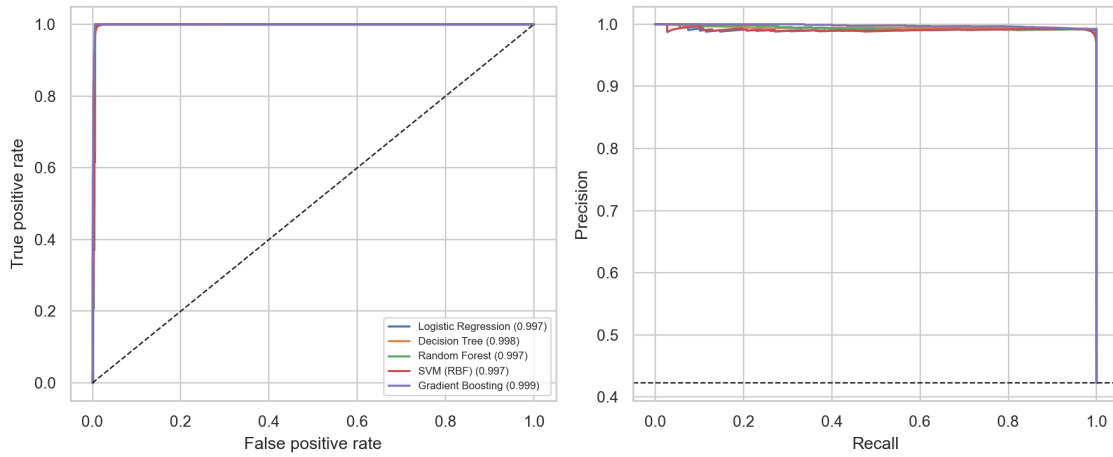


Figure 4: ROC (left) and precision-recall (right) curves on the test set. The dashed line on the right is the prevalence baseline.

		Logistic Regression		Decision Tree		Random Forest		SVM (RBF)		Gradient Boosting	
Actual	Normal	3833	49	3860	22	3860	22	3841	41	3858	24
	Susp.	2	2839	1	2840	1	2840	20	2821	0	2841
		Normal	Susp.	Normal	Susp.	Normal	Susp.	Normal	Susp.	Normal	Susp.
		Predicted		Predicted		Predicted		Predicted		Predicted	

Figure 5: Confusion matrices (rows: actual; columns: predicted).

6.3 Cross-validation and statistical significance

Table 5 confirms that the hold-out ranking is stable across folds. The standard deviations are at most 0.0014 F1.

Exact McNemar tests (Table 6) show that the Decision Tree is significantly better than the SVM and Logistic Regression ($p < 0.001$). Where the Decision Tree and the SVM disagree, the tree is right 39 times and the SVM once. The Decision Tree is *not* distinguishable from Random Forest, whose predictions are identical, or from Gradient Boosting ($p = 1.0$).

Table 5: 5-fold CV on the training set (mean $\pm$ sd).

Model	F1	ROC-AUC
Decision Tree	0.9963 $\pm$ 0.0012	0.9980 $\pm$ 0.0007
Random Forest	0.9963 $\pm$ 0.0012	0.9972 $\pm$ 0.0009
Gradient Boosting	0.9952 $\pm$ 0.0010	0.9983 $\pm$ 0.0005
Logistic Reg.	0.9900 $\pm$ 0.0014	0.9971 $\pm$ 0.0009
SVM (RBF)	0.9874 $\pm$ 0.0006	0.9971 $\pm$ 0.0009

Table 6: McNemar tests vs Decision Tree.

Other	DT only	Other only	p
Log. Reg.	29	1	<0.001
Random Forest	0	0	1.00
SVM (RBF)	39	1	<0.001
Grad. Boost.	2	1	1.00

6.4 What drives the performance? Rule baseline and ablation

Scores near 99.6% F1 call for scrutiny. A two-condition rule, learnt from the training data, reaches F1 0.9951 on the test set, only 0.0009 below the best model:

*flag if the description has any suspicious history **and** amount > 197,827*

It never misses a suspicious transaction and raises only 28 false alarms. The labels in this dataset therefore follow a nearly deterministic pattern: *large transactions in risk-prone merchant categories*. The ML models learn that pattern with slightly fewer false positives.

The feature-group ablation with Random Forest (Table 7) confirms this. Merchant/channel information and amount are both necessary, and neither is sufficient on its own:

- Removing description/channel drops F1 from 0.996 to 0.800.
- Amount and balance alone reach only 0.745.
- Description/channel alone reaches 0.839. Recall is perfect, but precision is poor.
- Calendar features add nothing (F1 unchanged at 0.996), despite the visible drift in the suspicious rate over time.

Table 7: Feature-group ablation (Random Forest, same split).

Feature set	Precision	Recall	F1	ROC-AUC
All features	0.9923	0.9996	0.9960	0.9971
Without date features	0.9923	0.9996	0.9960	0.9976
Description / channel only	0.7223	1.0000	0.8388	0.8615
Without description / channel	0.8160	0.7849	0.8001	0.8904
Amount & balance only	0.6020	0.9761	0.7447	0.7718

6.5 Class-imbalance handling

Random Forest gives the same F1 of 0.9960 with no re-balancing, with balanced class weights and with SMOTE. PR-AUC ranges only from 0.9923 to 0.9939. At a 42:58 class ratio, re-balancing is not needed. This contrasts with real card-fraud data, where fraud is often well below 1% of transactions and re-balancing matters a lot (Zhang, 2025; Ali et al., 2022).

6.6 Temporal robustness

In production, a model is trained on the past and used on the future. The models were therefore retrained on transactions before 17 March 2024 and tested on all later ones. The later period has a much lower suspicious rate, 19.6%.

The tree models and gradient boosting hold up fully: F1 is 0.999 to 1.000. The SVM with calendar features degrades sharply: F1 falls to 0.855 and recall to 0.750. Once calendar features are removed, its F1 recovers to 0.994.

Calendar features are a liability for drift-sensitive models. Year and month encode the historic base rate, and that rate changes over time. For deployment, they should be left out or monitored.

6.7 Generalisation to unseen merchants

The hardest test holds out whole merchant descriptions, using 5-fold **GroupKFold**. The model must then judge transactions at merchants it has never seen, using only channel, amount, balance and date. Table 8 shows that F1 falls to 0.86 to 0.88 and varies more across folds (sd $\approx$ 0.05). The ranking quality stays good, with ROC-AUC 0.95 to 0.97.

Gradient boosting and Random Forest generalise best. This is the most realistic estimate of how the models would handle a new e-commerce or crypto platform, and it shows why continuous retraining matters.

Table 8: Unseen-merchant generalisation (GroupKFold over 43 descriptions, 5 folds, mean).

Model	Precision	Recall	F1 ($\pm$ sd)	ROC-AUC
Gradient Boosting	0.9018	0.8618	0.8776 $\pm$ 0.053	0.9741
Random Forest	0.8778	0.8677	0.8694 $\pm$ 0.050	0.9735
Decision Tree	0.8436	0.8787	0.8584 $\pm$ 0.046	0.9629
SVM (RBF)	0.8416	0.8800	0.8573 $\pm$ 0.040	0.9528

6.8 Explainability

Figure 6 shows the global explanation of the Random Forest. Transaction amount (raw and log-scaled) is the most important feature, followed by the NETTXN and CRYPTO-EXCHANGE channel indicators. SHAP values show the direction of each effect:

- high amounts and crypto or online-payment channels push towards *suspicious*;
- IMPS transfers, salary credits and local-shop card payments push towards *normal*.

These explanations match the rule baseline. A compliance officer could check them easily, which is what regulators expect (European Parliament and Council of the European Union, 2024).

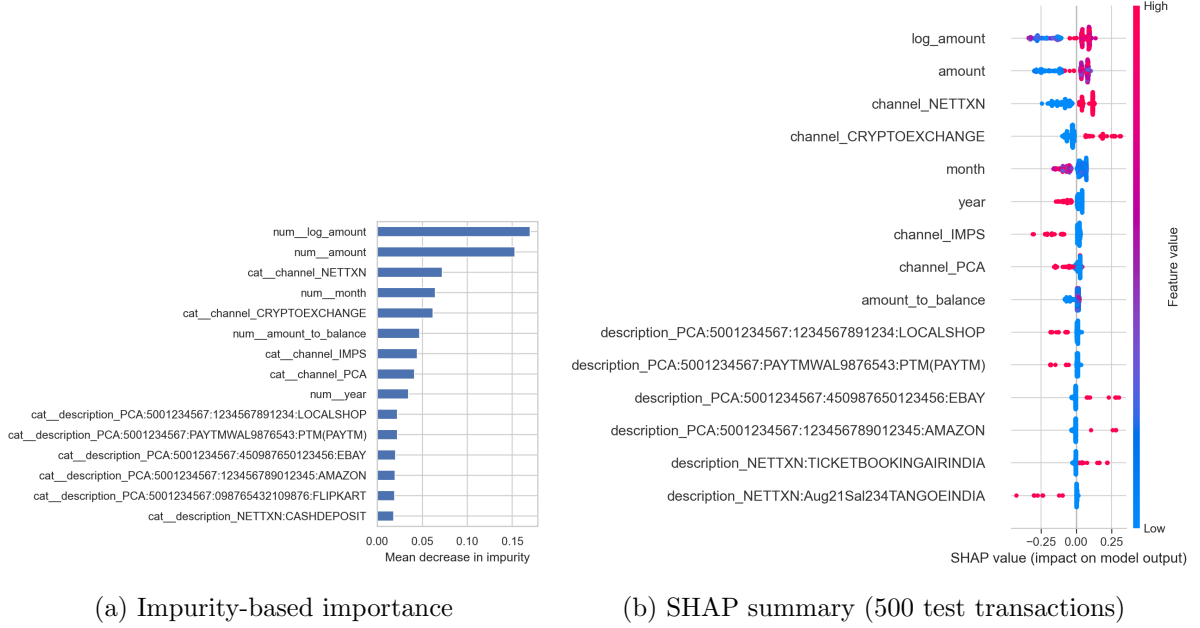


Figure 6: Global explanations of the Random Forest.

6.9 Operational trade-offs: threshold, latency and scalability

For the Decision Tree, precision and recall stay flat across decision thresholds from 0.10 to 0.65 (Figure 7a). The model is very confident, so the alert volume, about 426 per 1,000 transactions in this dataset, is set by the data rather than by the threshold.

Training time (Figure 7b) grows roughly linearly for the tree models: 0.09s for the Decision Tree and 0.42s for the Random Forest on 26,888 rows. For the SVM it grows super-linearly: 0.2s at 10% of the data, but 7.4s at 100%, a 35-fold increase for 10 times the data. Scoring latency is below 0.01 ms per transaction for the tree models and about 0.07 ms for the SVM.

At bank scale, with millions of transactions a day, these costs favour tree ensembles. Kernel SVMs would need approximation or distributed training.

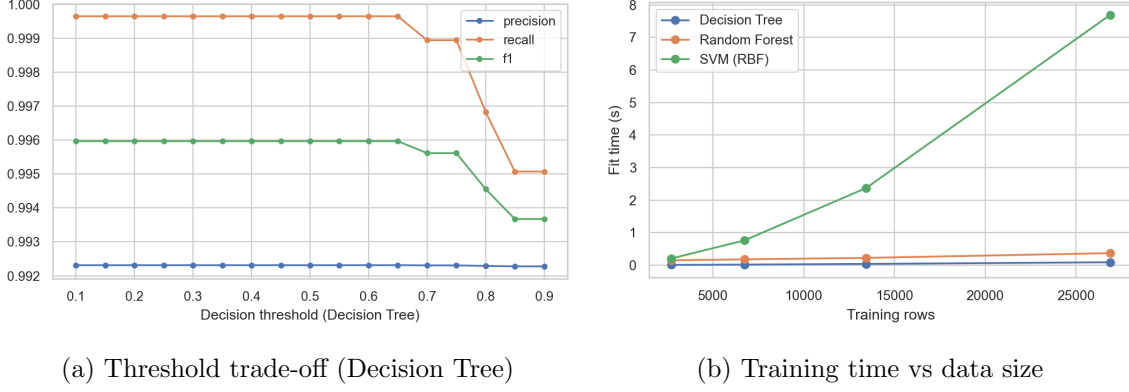


Figure 7: Operational characteristics.

7 Discussion

7.1 RQ1: How is BDA used to detect and prevent cyber threats in banking?

The literature and the experiments point to three complementary uses:

- Pattern discovery.* Analysing all transactions, rather than samples, revealed that risk is concentrated in specific channels (crypto and online payments) and at high amounts (Figure 2).
- Automated scoring.* Supervised models score each transaction in microseconds and flag it for review. Prevention comes from holding or stepping up authentication on a transaction before it settles.
- Intelligence and feedback.* Analyst decisions become new labels, closing the loop described by Montasari et al. (2020) and shown in Figure 1.

Irish banks already run such capabilities in their fraud and SOC functions. This study provides a transparent, reproducible template for them.

7.2 RQ2: How effective are ML and predictive analytics?

On this dataset they are very effective:

- Decision Tree and Random Forest reach F1 0.996 and recall 0.9996, significantly better than the SVM and Logistic Regression;
- gradient boosting reaches the best ROC-AUC;
- the tree models are also the fastest.

This agrees with the reviews that favour tree-based models for tabular fraud data (Ali et al., 2022; Naeem et al., 2026).

The robustness analysis also shows the limits. A simple rule comes within 0.001 F1 of the best model, and performance drops to about 0.87 F1 on unseen merchants. Headline accuracies above 99%, common in the literature, should therefore be read as properties

of the dataset as much as of the algorithm (Ring et al., 2019). The real value of ML here lies in three things:

- slightly fewer false alarms than the rule (22 against 28);
- smoother generalisation to new merchants than a lookup rule, which by construction cannot flag an unseen merchant at all;
- automatic re-learning as patterns change.

7.3 RQ3: What are the implementation challenges?

The experiments and the literature point to the following challenges:

- **Data quality and representativeness.** Public data is synthetic and simplified. Real data is siloed and subject to GDPR (Hasan et al., 2023; European Parliament and Council of the European Union, 2016).
- **Concept drift.** The suspicious rate moved from 50% in 2023 to 20% in 2024. Drift-sensitive models (the SVM with calendar features) lost 14 F1 points.
- **Generalisation to new attack surfaces.** Unseen merchants reduced F1 by about 13 points.
- **Explainability and accountability.** Regulators and customers need reasons for blocked payments, and the EU AI Act (European Parliament and Council of the European Union, 2024) adds transparency obligations.
- **Scalability.** Kernel methods scale poorly. Big-data platforms such as Spark (Zaharia et al., 2016) and skills to run them are needed.
- **Adversaries adapt.** Attackers adapt to detectors (Padmanaban et al., 2026), and static test sets do not show this.
- **Class imbalance in real fraud.** Real fraud is far rarer than in this dataset (Zhang, 2025).

7.4 RQ4: Recommended strategies for Irish banks

1. **Start interpretable, then ensemble.** Deploy tree-based models, whose explanations regulators can audit (SHAP and extracted rules). Keep the analyst rule as a benchmark and a fallback control.
2. **Evaluate as you deploy.** Report temporal and unseen-segment performance, not just random-split accuracy. Monitor drift, retrain on a schedule, and avoid features that only encode historic base rates.
3. **Build a governed data platform.** Use streaming ingestion with feature stores on Spark or Kafka, and role-based access. Run a Data Protection Impact Assessment under GDPR Art. 35. Align ICT-risk testing and incident reporting with DORA (European Parliament and Council of the European Union, 2022).
4. **Collaborate without sharing raw data.** Federated learning (Laddi et al., 2025) lets Irish banks pool fraud intelligence while keeping data local.
5. **Close the loop.** Feed analyst outcomes back as labels, and red-team the models with adversarial scenarios (Padmanaban et al., 2026).
6. **Invest in skills.** Build data-engineering, ML-ops and model-risk-management capacity. Hasan et al. identify gaps in these areas as a key barrier (Hasan et al., 2023).

7.5 Limitations

- **The dataset is synthetic in style and not Irish.** Merchant names and channels (e.g. IMPS) suggest an Indian retail-banking template. The findings therefore say how BDA *would* work for an Irish bank; they are not a measurement of Irish banks' actual performance.
- **The labels are almost deterministic.** Absolute scores overstate real-world performance, as shown in Section 6.4.
- **No cluster deployment was run.** A Java runtime for Spark was not available in the project environment. Scalability was therefore assessed on a single node with multi-core parallelism.
- **No adversarial evaluation or customer-level features.** Features such as velocity and device fingerprinting were not possible with six raw fields.

8 Conclusion and Future Work

This project analysed the role of Big Data Analytics in detecting and preventing cyber threats in banking. It built a reproducible experimental pipeline on 33,611 labelled bank transactions.

Decision Tree and Random Forest were the most effective and efficient of the proposed models (F1 0.996, recall 0.9996). They were significantly better than the SVM, which was also the least scalable. Gradient boosting matched them and gave the best ranking quality.

The robustness experiments are the main academic contribution. They show that the near-perfect scores come from a simple, explainable pattern: large transactions in crypto and online-payment channels. They also show that performance is much lower for unseen merchants, and that calendar features create drift risk.

For Irish banks, BDA delivers most when it is used to *discover*, *explain* and *operationalise* risk patterns under strong governance, rather than as a black-box accuracy engine.

Future work should:

- (i) validate the pipeline on real, highly imbalanced Irish or EU transaction data under a data-sharing agreement;
- (ii) deploy it on Spark Structured Streaming and measure end-to-end latency;
- (iii) add customer-level behavioural features and graph features linking accounts;
- (iv) evaluate adversarial robustness and federated training across institutions.

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A Reproducibility Guide

1. Install Python 3.10+ and the required packages:

```
pip install pandas numpy scikit-learn imbalanced-learn  
statsmodels shap matplotlib seaborn kaggle.
```
2. Put a Kaggle API token in `~/.kaggle/` (never commit it). Then run:

```
kaggle datasets download -d  
charanmaik/bank-transaction-records-along-suspicious-flags --unzip -p data.
```
3. Run `python src/pipeline.py`, which takes about four minutes. All tables are written to `results/` and all figures to `figures/`.
4. Optionally, open `notebooks/analysis.ipynb` to view the results interactively.
5. Build this report with `cd report && latexmk -pdf thesis.tex`.

B Project Plan

Figure 8 shows the original project schedule. All phases (literature review, data collection, pre-processing, model development, evaluation, analysis and report writing) have been completed.

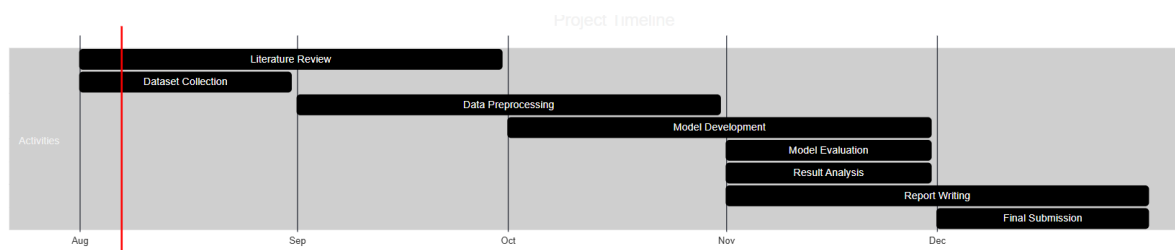


Figure 8: Project Gantt chart.